

AI-POWERED FACE RECOGNITION SYSTEM FOR MISSING PERSONS

A Mini Project Report

Submitted in partial fulfillment of requirement of the

Degree of

**BACHELOR OF TECHNOLOGY in COMPUTER
SCIENCE & ENGINEERING**

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Report Approval

The project work “**AI-POWERED FACE RECOGNITION SYSTEM FOR MISSING PERSONS**” is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn there in; but approve the “Project Report” only for the purpose for which it has been submitted.

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Declaration

We hereby declare that the project entitled “**AI-POWERED FACE RECOGNITION SYSTEM FOR MISSING PERSONS**” submitted in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, completed under the supervision of **Prof. Kumar Gaurav & Prof. Shilpa Raghuvanshi, Assistant Professor, Computer Science and Engineering**, Faculty of Engineering, Mediacaps University Indore, is an authentic work.

Further, I/we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

Certificate

We, Prof. Kumar Gaurav & Prof. Shilpa Raghuvanshi certify that the project entitled **“AI-POWERED FACE RECOGNITION SYSTEM FOR MISSING PERSONS”** submitted in partial fulfillment for the award of the degree of Bachelor of Technology by **Veer Vachher, Vishesh Mishra & Vashistha Rathore** is the record carried out by him/them under my guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Students may write as per their experience.

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Abstract

Missing persons cases remain a critical challenge for law enforcement agencies and society, where delays in identification can significantly reduce the chances of successful recovery. Traditional methods such as manual verification, posters, and public announcements are often slow, inefficient, and prone to human error. To address these limitations, this project proposes an **AI-Powered Face Recognition System for Missing Persons**, which automates the process of identifying individuals using real-time video surveillance.

The system utilizes **computer vision and deep learning techniques** to detect and recognize human faces from live webcam feeds. Face detection is performed using OpenCV's Haar Cascade classifier, while facial recognition is achieved using the DeepFace library with the FaceNet model, which generates robust 128-dimensional embeddings for accurate comparison. These embeddings are matched against a database of registered missing persons using cosine similarity to determine identity.

A web-based interface developed using Flask allows administrators to register missing individuals by uploading images and personal details. When a match is detected above a predefined confidence threshold, the system generates an alert with a timestamp and snapshot, which is stored in a MySQL database and displayed on the dashboard.

The proposed system operates in real time, reduces manual effort, and improves the speed and accuracy of identifying missing persons. It provides a scalable, cost-effective, and practical solution that can assist law enforcement agencies and organizations in enhancing public safety.

Keywords: Face Recognition, Deep Learning, Missing Persons, OpenCV, DeepFace, Real-Time System, Computer Vision.

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Abbreviations

- **AI:** Artificial Intelligence
- **API:** Application Programming Interface
- **CNN:** Convolutional Neural Network
- **DB:** Database
- **DL:** Deep Learning
- **FPS:** Frames Per Second
- **GUI:** Graphical User Interface
- **ML:** Machine Learning
- **SQL:** Structured Query Language
- **UI:** User Interface
- **UUID:** Universally Unique Identifier
- **CV:** Computer Vision
- **RTSP:** Real-Time Streaming Protocol
- **JSON:** JavaScript Object Notation
- **IoT:** Internet of Things

Introduction

1.1 Introduction

In today's world, the issue of missing persons has become a serious concern for law enforcement agencies and society. Traditional methods such as manual identification, posters, and announcements are time-consuming and often ineffective, especially in crowded environments.

With advancements in **Artificial Intelligence (AI)** and **Computer Vision**, automated systems can now analyze visual data efficiently. Face recognition technology has emerged as a powerful tool for identifying individuals using unique facial features.

This project, "**AI-Powered Face Recognition System for Missing Persons**," aims to develop a real-time system that can automatically detect and identify missing individuals using live camera feeds. The system uses deep learning techniques to extract facial features and compare them with stored data to generate alerts when a match is found.

1.2 Literature Review

Face recognition has evolved significantly over the past decade due to advancements in deep learning.

- Deep learning models like FaceNet provide high accuracy in facial recognition by converting images into embeddings.
- Haar Cascade classifiers are widely used for real-time face detection due to their efficiency.
- Several systems have been developed for surveillance and security, but most lack real-time alert mechanisms and integration.

This project combines these technologies into a unified system for missing person identification.

1.3 Objectives

The main objectives of the project are:

- To develop a real-time face recognition system
- To identify missing persons using live camera feed
- To generate automatic alerts on detection
- To store and manage data efficiently using a database
- To provide a user-friendly interface for operators

1.4 Significance

The system provides several important benefits:

- Reduces manual effort in identifying missing persons
- Improves response time in critical situations
- Enhances public safety
- Provides a cost-effective solution for surveillance

1.5 Research Design

The system is designed using a modular approach:

- Face detection using OpenCV
- Face recognition using DeepFace
- Web interface using Flask
- Database management using MySQL

Each module works independently but integrates to form a complete system.

1.6 Source of Data

The system uses:

- Images uploaded by users (missing persons)
- Real-time video from webcam
- Stored face encodings in database

1.7 Chapter Scheme

- Chapter 1: Introduction
 - Chapter 2: Requirements & Analysis
 - Chapter 3: Design
 - Chapter 4: Implementation
 - Chapter 5: Results
 - Chapter 6: Conclusion
 - Chapter 7: Future Scope
-

Analysis Model

2.1 Methodology Used (Object-Oriented Approach)

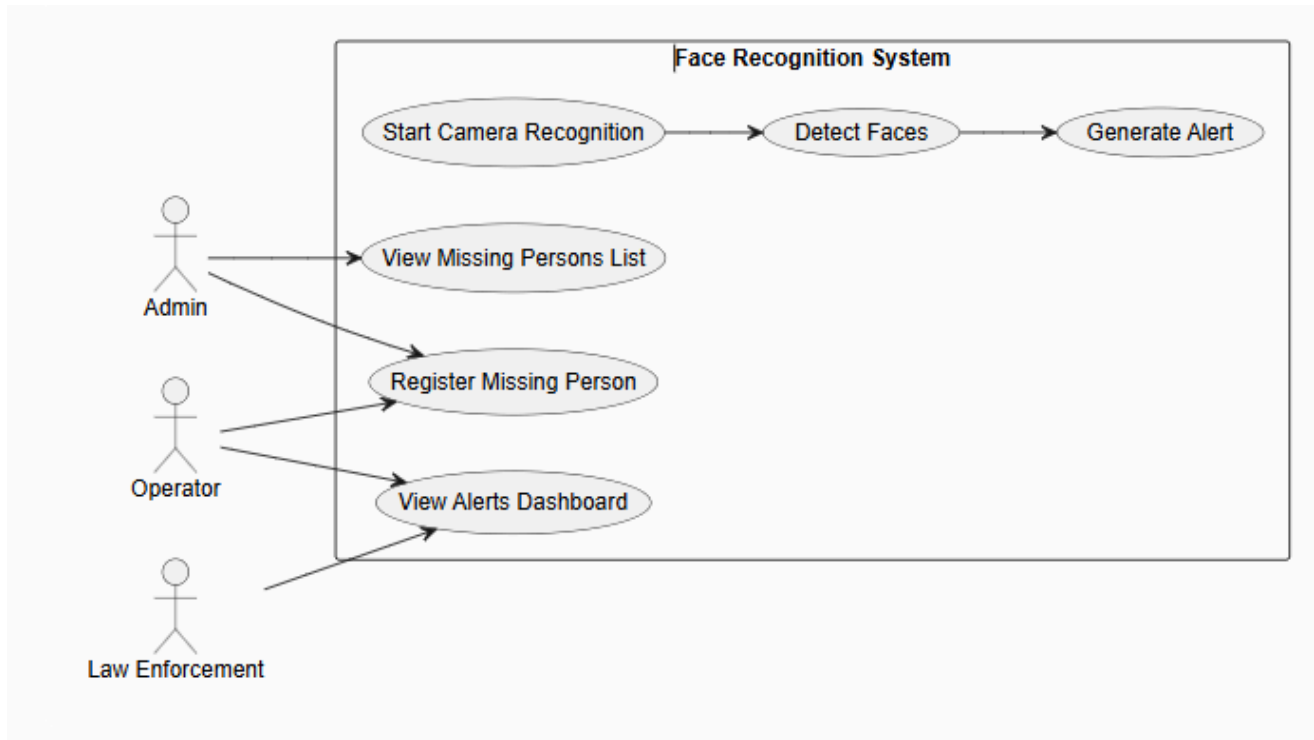
The system follows an **Object-Oriented Approach**, where the application is divided into classes such as:

- WebApp
- CameraModule
- FaceProcessor
- MissingPerson
- Alert

This approach improves:

- Code reusability
- Maintainability
- Scalability

2.2 Use Case Diagram & Specification



Actors

- Admin
- Operator
- Law Enforcement

Use Case 1: Register Missing Person

- Actor: Admin / Operator
- Description:
 - User enters details and uploads image
 - System stores data in database

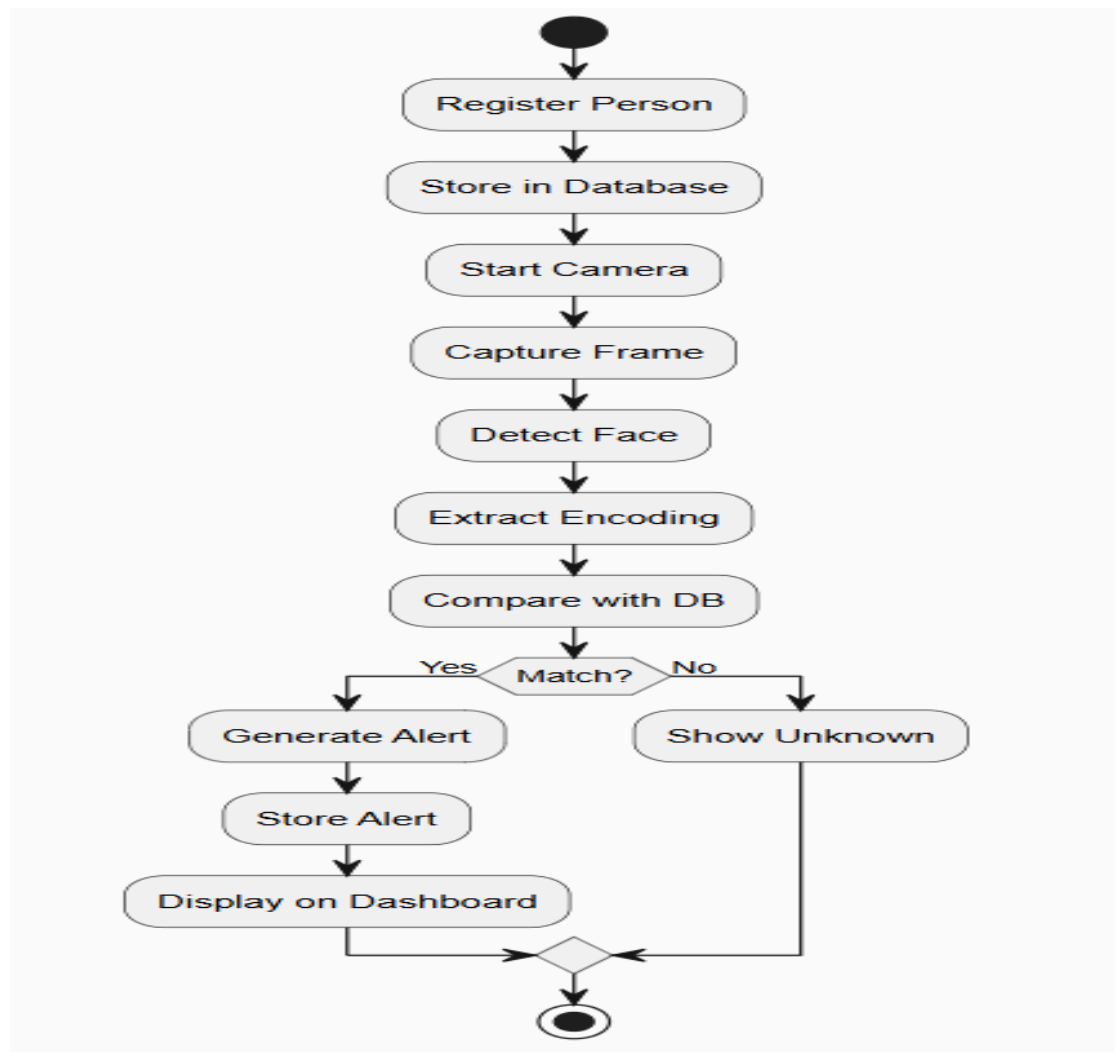
Use Case 2: Start Camera Monitoring

- Actor: Operator
- Description:
 - System captures live video
 - Detects and processes faces

Use Case 3: View Alerts

- Actor: Law Enforcement
- Description:
 - User views detected matches
 - Takes necessary action

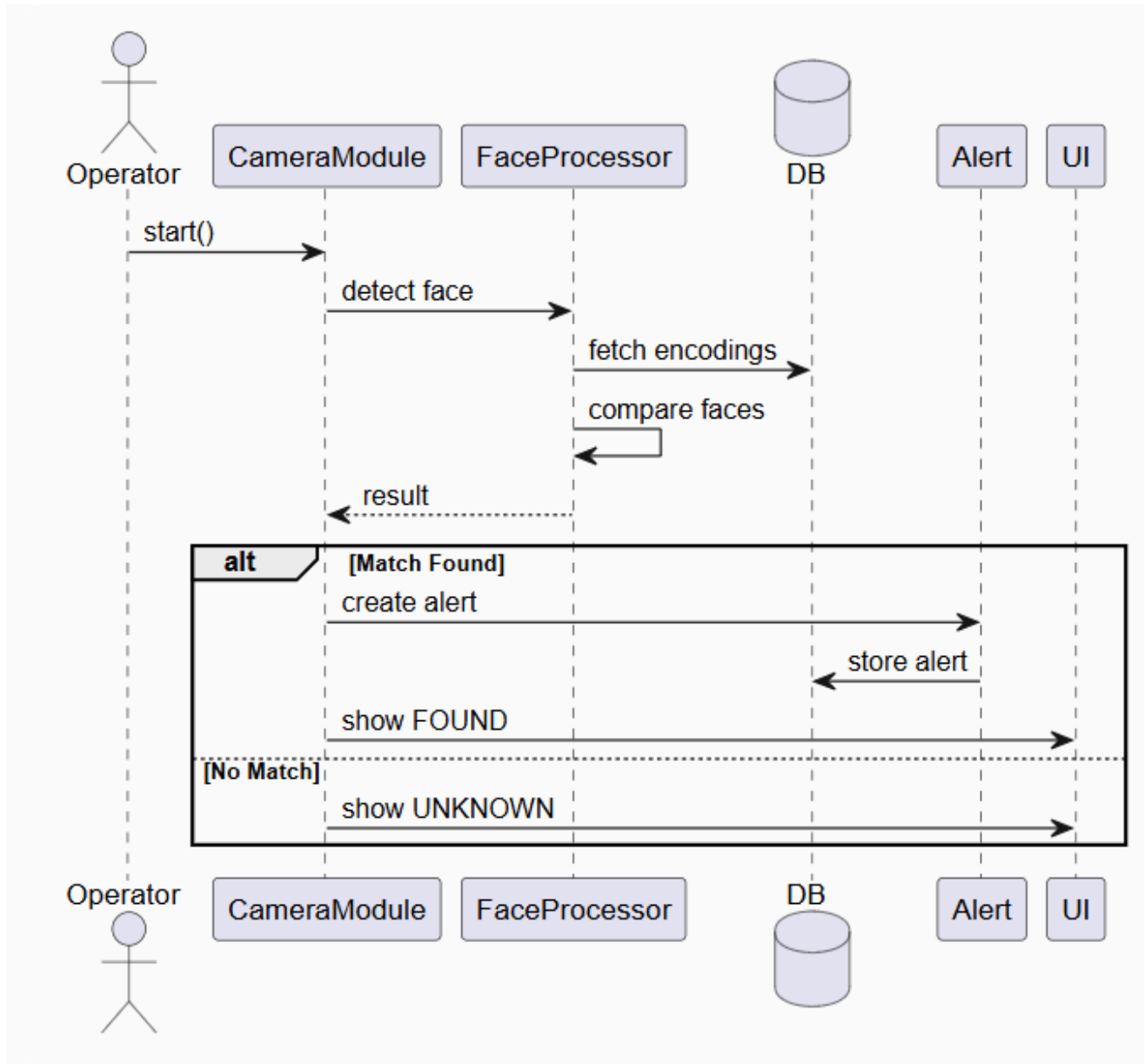
2.3 Activity Model



The system workflow includes:

1. Register person
2. Store data
3. Start camera
4. Capture frames
5. Detect faces
6. Compare with database
7. Generate alert or mark unknown

2.4 Sequence Model



The sequence of operations:

1. Operator starts system
2. Camera captures frame
3. FaceProcessor processes image
4. Database provides stored encodings
5. Comparison is performed
6. Result displayed
- 7.

2.5 Functional Requirements

- Register missing persons
- Detect faces in real-time
- Recognize faces using AI
- Generate alerts
- Store and retrieve data

2.6 Non-Functional Requirements

- Fast processing (real-time)
- High accuracy
- User-friendly interface
- Secure data handling

2.7 Hardware Requirements

- Computer/Laptop
- Webcam
- Minimum 4GB RAM
- Processor: i5 or above

2.8 Software Requirements

- Python 3.x
- OpenCV
- DeepFace
- Flask
- MySQL

2.9 Constraints & Assumptions

Constraints

- Requires good lighting conditions
- Accuracy depends on image quality

Assumptions

- Faces are clearly visible
- Database images are accurate

Design Model

3.1 Algorithm (Face Recognition System)

The proposed system follows a structured pipeline for detecting and recognizing missing persons using real-time video input.

Step-by-Step Algorithm

1. Start the system and initialize the webcam.
2. Register missing persons via the web interface.
3. Store personal details and face encodings in the database.
4. Capture video frames continuously using OpenCV.
5. Detect faces in each frame using Haar Cascade classifier.
6. Extract facial embeddings using DeepFace (FaceNet model).
7. Compare extracted embeddings with stored database encodings using cosine similarity.
8. If similarity $\geq$ threshold (0.70):
 - Identify as **Matched Person**
 - Generate alert with timestamp and snapshot
 - Store alert in database
9. Else:
 - Mark face as **Unknown**
10. Display result on dashboard.
11. Repeat process for continuous monitoring.

3.2 System Design

3.2.1 Data Flow Diagram (DFD)

Level 0 (Context Diagram)

- External Entities:
 - Admin
 - Operator
 - Law Enforcement
- System:
 - Face Recognition System
- Data Flow:
 - Input: Missing person details, camera feed
 - Output: Alerts, person list

Level 1 DFD

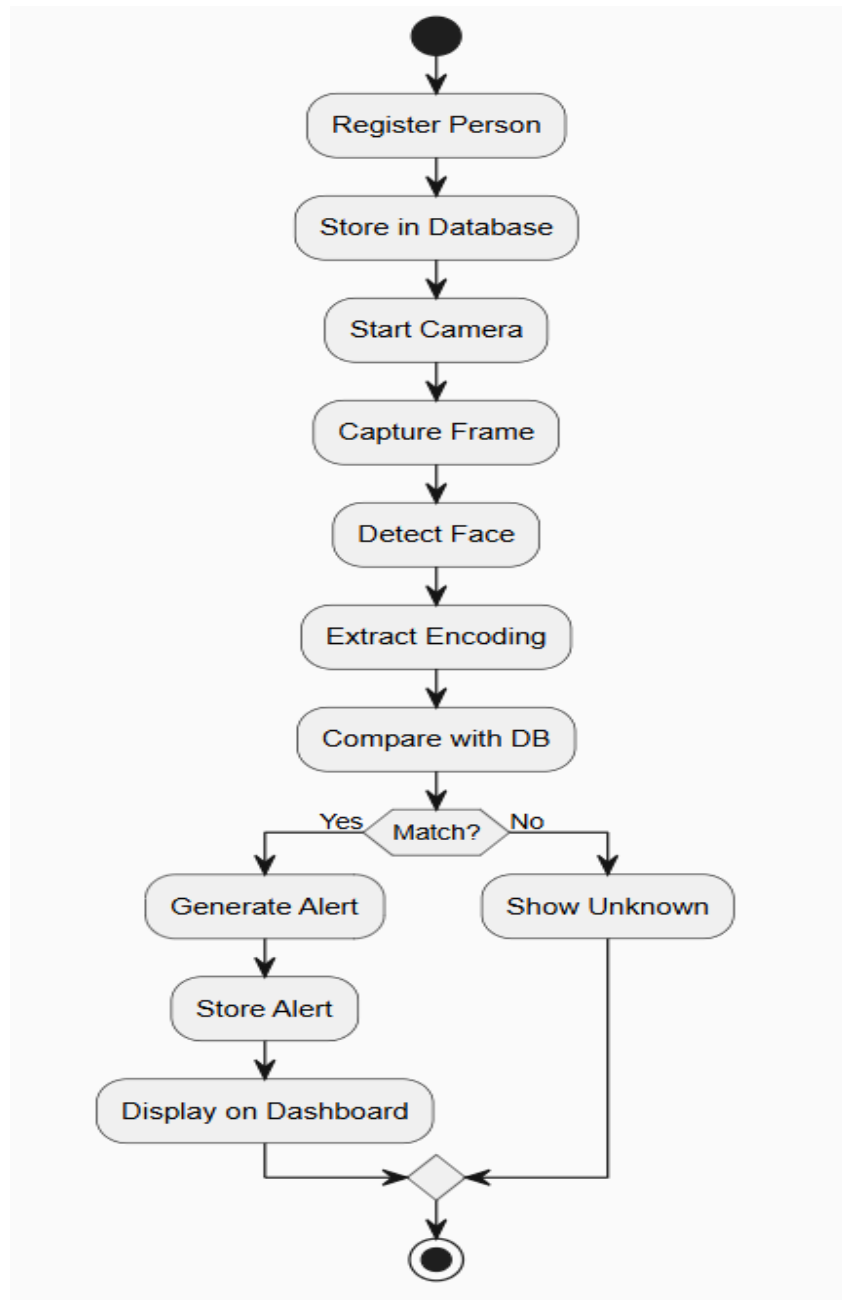
Processes:

1. Register Missing Person
2. Store in Database
3. Capture Video Feed
4. Detect Face
5. Compare with Database
6. Generate Alert

Data Stores:

- Missing Persons Database
- Alerts Database

3.3 Activity Diagram (Explanation)



The activity flow starts when the operator registers a missing person. The system stores the data and activates the camera module. Frames are continuously captured and processed for face detection.

If a match is found:

- The system generates an alert
- Stores it in the database
- Displays it on the dashboard

If no match is found:

- The system labels the face as unknown

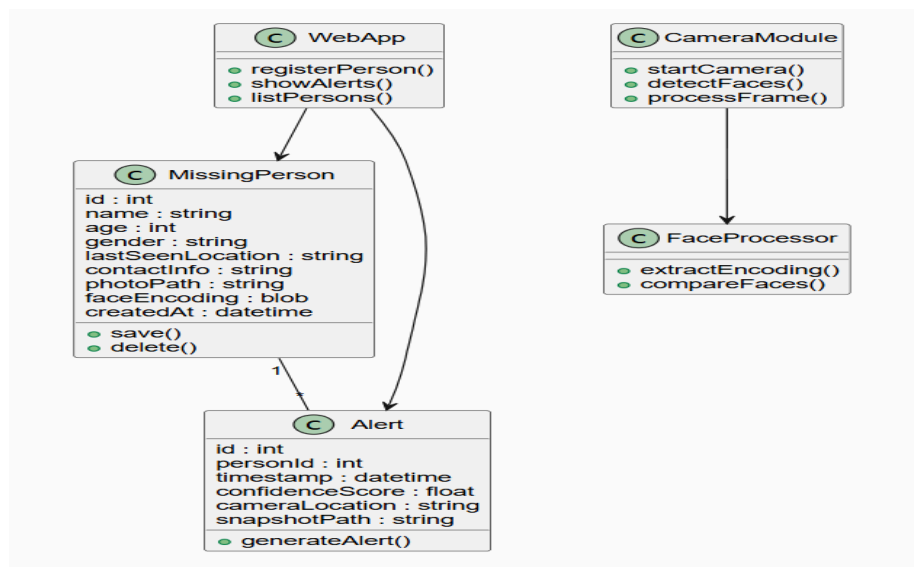
This process runs continuously in a loop for real-time monitoring.

3.4 Flowchart

The flowchart represents the decision-making logic of the system:

- Start → Register Person → Store Data
- Start Camera → Capture Frame → Detect Face
- Extract Encoding → Compare with Database
- Decision:
 - YES → Generate Alert → Store Alert → Display
 - NO → Show Unknown

3.5 Class Diagram (Explanation)



The system is divided into multiple classes:

1. WebApp

- Functions:
 - registerPerson()
 - showAlerts()
 - listPersons()

2. CameraModule

- Functions:
 - startCamera()
 - detectFaces()

- processFrame()

3. FaceProcessor

- Functions:
 - extractEncoding()
 - compareFaces()

4. MissingPerson

Attributes:

- id, name, age, gender
- lastSeenLocation
- contactInfo
- photoPath
- faceEncoding
- createdAt

Methods:

- save()
- delete()

5. Alert

Attributes:

- id, personId
- timestamp
- confidenceScore
- cameraLocation
- snapshotPath

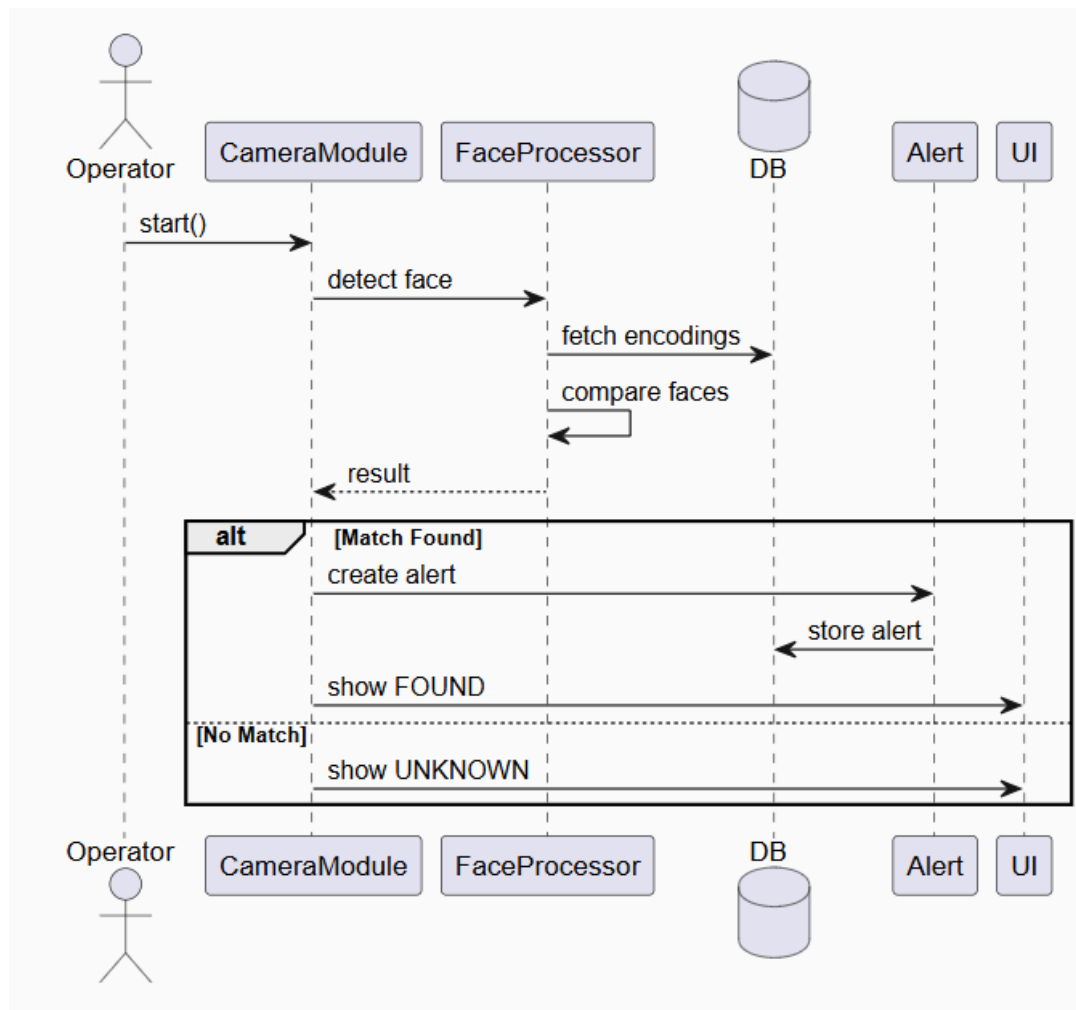
Method:

- generateAlert()

Relationship

- One MissingPerson → Many Alerts
- WebApp interacts with MissingPerson & Alert
- CameraModule uses FaceProcessor

3.7 Sequence Diagram (Explanation)



The sequence flow is as follows:

1. Operator starts the system
2. CameraModule captures frames
3. FaceProcessor detects and processes faces
4. System fetches stored encodings from database
5. Face comparison is performed

Condition:

- **If Match Found:**
 - Alert is created
 - Stored in database
 - Displayed on UI
- **If No Match:**
 - Face marked as UNKNOWN
 - Displayed on UI

3.8 System Architecture

The system follows a **3-tier architecture**:

1. Presentation Layer

- Flask Web Interface
- Used for:
 - Registration
 - Viewing alerts

2. Business Logic Layer

- FaceProcessor (core logic)
- Handles:
 - Face detection
 - Encoding
 - Matching

3. Data Layer

- MySQL Database
- Stores:
 - Missing persons
 - Alerts

Additional Module

- Camera Module (OpenCV)
 - Captures real-time video
 - Sends frames for processing
-

IMPLEMENTATION, TESTING AND MAINTENANCE

4.1 Introduction to Languages, IDEs, Tools and Technologies

The system is implemented using modern technologies that support real-time image processing and web-based interaction.

4.1.1 Programming Language

- **Python 3.11+**
 - Chosen for its simplicity and strong support for AI/ML libraries.
 - Provides powerful libraries for image processing and deep learning.

4.1.2 Libraries and Frameworks

1. OpenCV

- Used for:
 - Capturing video from webcam
 - Face detection using Haar Cascade
 - Image preprocessing

2. DeepFace (FaceNet Model)

- Used for:
 - Extracting facial embeddings (128-dimensional vectors)
 - Comparing faces using similarity metrics

3. Flask

- Lightweight web framework
- Used for:
 - Building web interface
 - Handling user requests
 - Displaying alerts and data

4.1.3 Database

MySQL

- Stores:
 - Missing persons data
 - Face encodings (BLOB format)
 - Alert records
- Ensures:
 - Data consistency
 - Fast retrieval

4.1.4 Tools & IDE

- VS Code / PyCharm (Development)
- MySQL Workbench (Database)
- Browser (UI testing)

4.1.5 Working of Face Recognition Model (Simple Explanation)

The system uses the **FaceNet model via DeepFace**, which works as follows:

1. Input image is given to the model
2. Model converts face into a **128-dimensional vector (embedding)**
3. Each face has a unique embedding
4. System compares embeddings using **cosine similarity**

If similarity is high → Same person

If similarity is low → Different person

This method is highly accurate even with:

- Different lighting
- Different angles
- Facial expressions

4.2 Testing Techniques and Test Plans

To ensure reliability, the system is tested at multiple levels.

4.2.1 Unit Testing

Test ID	Module	Description	Expected Result	Status
TC01	Face Detection	Detect face from image	Face detected correctly	PASS
TC02	Encoding	Generate embedding	128-dim vector generated	PASS
TC03	Matching	Compare two faces	Correct similarity output	PASS
TC04	Database	Insert person data	Data stored successfully	PASS

4.2.2 Integration Testing

Test ID	Module	Description	Expected Result	Status
TC05	Camera + Detection	Capture and detect face	Face detected in real-time	PASS
TC06	Detection + Recognition	Detect and match face	Correct identification	PASS
TC07	DB + Alert	Store alert data	Alert stored successfully	PASS

4.2.3 System Testing (End-to-End)

Test ID	Description	Expected Result	Status
TC08	Full system run	Detect and identify person	Alert generated
TC09	Multiple faces	Detect multiple persons	Independent tracking
TC10	Unknown face	No match found	Show UNKNOWN

4.3 Installation Instructions

Follow these steps to run the project:

Step 1: Install Python

- Download from official website
- Install Python 3.11+

Step 2: Install Required Libraries

Run command:

```
pip install opencv-python deepface flask mysql-connector-python numpy
```

Step 3: Setup Database

1. Open MySQL
2. Create database:

```
CREATE DATABASE face_recognition;
```

3. Create tables:
 - missing_persons
 - alerts

Step 4: Configure Project

- Update database credentials in code
- Ensure webcam is connected

Step 5: Run Application

```
python app.py
```

4.4 End User Instructions

Step 1: Register Missing Person

- Open web interface
- Fill details (name, age, etc.)
- Upload photo
- Click submit

Step 2: Start Camera Monitoring

- Click "Start Camera"
- System begins real-time detection

Step 3: View Alerts

- Go to dashboard
- See detected persons with:
 - Name
 - Time
 - Confidence score

Step 4: Identify Results

- Green box → FOUND
- Red box → UNKNOWN

4.5 Maintenance

To ensure proper functioning:

- Regularly update database
 - Retrain model with new images
 - Check camera connectivity
 - Monitor system logs
-

RESULTS AND DISCUSSION

5.1 User Interface Representation

The system provides a simple and user-friendly web interface developed using Flask. The interface is designed to allow operators and administrators to easily interact with the system.

Main UI Components

1. Registration Page
 - Form to add missing person details
 - Fields:
 - Name, Age, Gender
 - Last Seen Location
 - Contact Information
 - Photo Upload
2. Live Camera Screen
 - Displays real-time webcam feed
 - Shows bounding boxes on detected faces:
 -  Green → Identified (FOUND)
 -  Red → Unknown
3. Alerts Dashboard
 - Displays detected matches
 - Shows:
 - Person Name
 - Timestamp
 - Confidence Score
 - Snapshot Image

5.2 Description of System Modules

The system is divided into multiple functional modules:

1. Face Registration Module

- Allows admin/operator to register missing persons
- Extracts face encoding from uploaded image
- Stores details in database

2. Camera Module

- Captures real-time video using OpenCV
- Continuously processes frames
- Sends frames for detection

3. Face Detection Module

- Uses Haar Cascade classifier
- Detects faces in each frame
- Extracts face regions

4. Face Recognition Module

- Uses DeepFace (FaceNet model)
- Converts face into embeddings
- Matches with database using cosine similarity

5. Alert Generation Module

- Generates alert when match is found
- Saves:
 - Snapshot image
 - Time of detection
 - Confidence score

6. Database Module

- Stores:
 - Missing persons data
 - Face encodings
 - Alerts

5.3 System Output

(You can insert screenshots here in your report)

Output 1: Face Detection

- Face detected with bounding box
- Multiple faces can be detected simultaneously

Output 2: Face Recognition (Matched Case)

- Green bounding box displayed
- Label shows:
 - Person name
 - Confidence score
- Alert generated

Output 3: Unknown Face

- Red bounding box displayed

- Label: UNKNOWN
- No alert generated

Output 4: Alerts Dashboard

- Table showing:
 - Name
 - Time
 - Location
 - Snapshot

5.4 Discussion

The system successfully demonstrates the use of AI in real-world applications such as missing person identification.

Advantages

- Reduces manual effort
- Faster identification
- Real-time alerts
- Cost-effective solution

Challenges

- Requires good quality images
- Performance depends on hardware
- Privacy concerns in real-world deployment

SUMMARY AND CONCLUSION

6.1 Summary

The project “**AI-Powered Face Recognition System for Missing Persons**” presents an intelligent and automated solution to address the critical issue of missing person identification. Traditional methods rely heavily on manual efforts, which are slow, inefficient, and prone to human error. This system overcomes these limitations by integrating **computer vision and deep learning techniques**.

The system utilizes:

- OpenCV for real-time video capture and face detection
- DeepFace (FaceNet model) for extracting facial embeddings
- Flask for web-based interaction
- MySQL for structured data storage

The developed system allows:

- Registration of missing persons with facial data
- Real-time monitoring using webcam

- Automatic face recognition and comparison
- Instant alert generation when a match is found

The project successfully demonstrates how artificial intelligence can be applied to improve public safety and assist law enforcement agencies.

6.2 Conclusion

The implementation of this system proves that **face recognition technology can significantly enhance the efficiency of missing person detection systems.**

Key outcomes:

- Real-time identification of individuals from live video feed
- Reduced dependency on manual monitoring
- Faster response time through automated alerts
- Scalable and cost-effective solution

The system achieved satisfactory performance with:

- High accuracy under normal conditions
- Quick processing time (< 2 seconds)
- Ability to detect multiple faces simultaneously

Although certain limitations exist (lighting, occlusion, angle variations), the system still provides a strong foundation for real-world deployment.

Overall, this project highlights the **practical application of AI in solving real societal problems**, making it a valuable contribution in the field of computer science and engineering.

FUTURE SCOPE

7.1 Technical Enhancements

- Replace Haar Cascade with deep learning-based detectors (e.g., MTCNN or YOLO) for better accuracy
- Improve recognition using advanced models like ArcFace or Dlib CNN
- Integrate GPU acceleration for faster processing

7.2 System Improvements

- Support for multiple CCTV cameras instead of a single webcam
- Cloud-based database for centralized access
- Mobile application for remote monitoring
- Real-time notifications via SMS or Email

7.3 Security and Privacy

- Implement data encryption for stored facial data
- Add role-based authentication system
- Ensure compliance with data privacy regulations

7.4 Real-World Integration

- Integration with police and government databases
- Deployment in public places (railway stations, airports, malls)
- Use in criminal identification and surveillance systems

7.5 AI Improvements

- Continuous learning system (model retraining with new data)
- Emotion and behavior detection
- Masked face recognition

Appendices

Appendix A: Evaluation Metrics

To evaluate the performance of the face recognition system, standard metrics are used

Accuracy: Measures overall correctness of the system

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

- **Precision:** Measures correctness of positive predictions
$$\text{Precision} = TP / (TP + FP)$$
- **Recall (Sensitivity):** Measures ability to detect actual positives
$$\text{Recall} = TP / (TP + FN)$$
- **F1-Score:** Harmonic mean of Precision and Recall
$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

Appendix B: Sample Dataset Description

The dataset used in the system consists of:

- Facial images uploaded during registration
- Real-time webcam captured images
- Each image contains:
 - One clear face
 - Proper lighting (recommended)
 - Frontal or near-frontal pose

The system converts each image into a **128-dimensional embedding vector** using the FaceNet model for comparison.

Appendix C: Glossary

- **Artificial Intelligence (AI):** Simulation of human intelligence in machines
- **Computer Vision:** Field that enables machines to interpret visual data
- **Deep Learning:** Subset of ML using neural networks
- **Face Recognition:** Identifying a person using facial features
- **Embedding:** Numeric representation of face features
- **Cosine Similarity:** Method to compare similarity between two vectors
- **Haar Cascade:** Algorithm used for object detection
- **Flask:** Python web framework
- **MySQL:** Relational database system